

Angular – Hands On 2

Data Binding, Lifecycle Hooks & Component Communication

Objective:

To implement data binding, lifecycle hooks, and parent-child communication in the Student Course Portal using Angular.

Home.ts:

```
import { Component, OnInit, OnDestroy } from '@angular/core';
import { FormsModule } from '@angular/forms';

@Component({
  selector: 'app-home',
  standalone: true,
  imports: [FormsModule],
  templateUrl: './home.html',
  styleUrls: ['./home.css']
})
export class Home implements OnInit, OnDestroy {
  // String Interpolation
  portalName = 'Student Course Portal';

  // Property Binding
  isPortalActive = true;

  // Event Binding
  message = '';

  // Two-Way Binding
  searchTerm = '';

  constructor() {}

  ngOnInit(): void {
    console.log('HomeComponent initialized - courses loaded');
  }

  ngOnDestroy(): void {
    console.log('HomeComponent destroyed');
  }
}
```

```
onEnrollClick(): void {  
    this.message = 'Enrollment opened!';  
}  
}
```

Home.html:

```
<h1>{{ portalName }}</h1>  
<p>Welcome to the Student Course Portal.</p>  
<button  
    [disabled]="!isPortalActive"  
    (click)="onEnrollClick()">  
    Enroll Now  
</button>  
<p>{{ message }}</p>  
<hr>  
<input  
    type="text"  
    placeholder="Search Courses"  
    [(ngModel)]="searchTerm">  
<p>Searching for: {{ searchTerm }}</p>
```

Home.css:

```
h1{  
    color: navy;  
}  
button{  
    padding:8px 15px;  
    margin-top:10px;  
}  
input{  
    margin-top:20px;  
    padding:8px;  
    width:250px;
```

```
}  
  
p{  
    font-size:18px;  
}
```

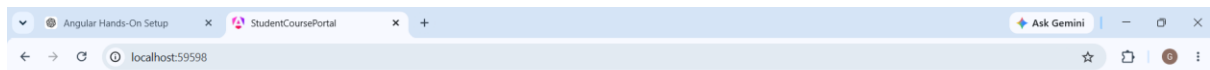
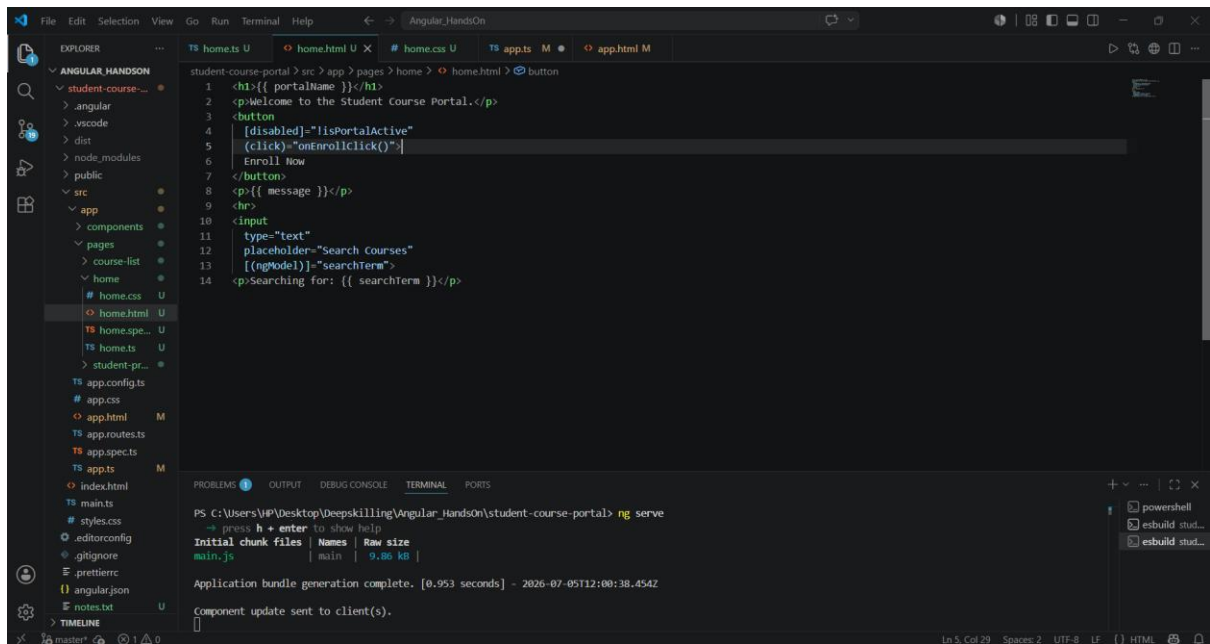
App.ts:

```
import { Component, signal } from '@angular/core';  
import { RouterOutlet } from '@angular/router';  
import { Header } from '../components/header/header';  
import { Home } from '../pages/home/home';  
  
@Component({  
    selector: 'app-root',  
    imports: [  
        Header,  
        Home,  
        RouterOutlet  
    ],  
    templateUrl: './app.html',  
    styleUrls: ['./app.css']  
})  
  
export class App {  
    protected readonly title = signal('student-course-portal');  
}
```

App.html:

```
<app-header></app-header>  
  
<app-home></app-home>  
  
<router-outlet></router-outlet>
```

Outputs:



Student Course Portal

[HomeCoursesProfile](#)

Student Course Portal

Welcome to the Student Course Portal.

Enroll Now

Enrollment opened!

Angular

Searching for: Angular